

# Validity Is Not Difficulty: Supplementary Material

Anonymous CVPR submission

Paper ID \*\*\*\*\*

## A. SceneIR Schema and Coordinate Conventions

SceneIR records explicit  $T_{\text{city} \leftarrow \text{ego}}$ ,  $T_{\text{ego} \leftarrow \text{sensor}}$ , and  $T_{\text{city} \leftarrow \text{actor}}$  transforms. A return  $x_{s,t}$  is fused in Actor coordinates as

$$x_i = T_{\text{city} \leftarrow \text{actor}, t}^{-1} T_{\text{city} \leftarrow \text{ego}, t} T_{\text{ego} \leftarrow \text{sensor}, t} x_{s,t}. \quad (1)$$

The physical layer stores Actor-local surfels, known/unknown masks, normals, temporal and view support, matched rays, and source timestamps. Appearance primitives are referenced only through attachment provenance; Gaussian opacity is never read as collision occupancy. Actor identity, category, extent, trajectory, lifecycle, and analytic hazard fields are immutable across compiler actions.

**Four actions.** KEEP preserves supported query evidence. PROJECT moves an artifact only to its matched observed termination. COMPLETE adds a canonical surfel only when temporal and view support satisfy the frozen contract. UNKNOWN is the fallback when evidence is insufficient; it never becomes free.

## B. Matched-Ray Physical Certificate

Automotive LiDAR is 2.5D ray evidence, not merely an unordered 3D point set [2]. For origin  $o$ , direction  $v$ , and observed termination  $d$ , the measured free interval is

$$\mathcal{F}(o, v, d) = \{o + sv \mid 0 \leq s < d - \epsilon\}. \quad (2)$$

A provenance-bearing PROJECT candidate is placed at the same ray's termination  $o + dv$ . It therefore cannot lie in  $\mathcal{F}$ . A candidate without an exact ray is marked UNKNOWN; nearest-surface projection is not substituted. This is the source of the paired zero-free-space result. It is a deterministic contract for matched rays, not a completeness proof for the whole Actor surface or a probabilistic sensor-noise guarantee.

The held-out targets are later frames excluded from both canonical fusion and query compilation. Depth, ray-termination, Chamfer, precision, and recall are computed

against these target-only returns. This separation explains why zero paired free-space violation can coexist with the observed 17.19% per-Actor Chamfer tail.

**Bidirectional visibility partition.** For each held-out target ray, the nearest output within the frozen 0.20 m beam tube is classified as early, hit, or late; otherwise it is unmatched. In the reverse direction, an output point is supported at the first return, contradicted if it lies in the observed free interval, occluded if it is behind the return, and unknown when off ray. Occluded and unknown points are not penalized as free-space violations, preventing unseen Actor backs from being conflated with measurable contradictions.

The pooled result does not imply Actor-wise dominance: 64.04% add no early/visible contradiction and only 1.58% have absolute zero visible contradictions. All-Actor output attribution assigns 95.12% of new early rays to COMPLETE, but each costs against 14.96 new matched hits; target early returns fall by 9,439 overall. Moreover, only 13.49% of contradicted output points are completion points, so source-ray carving cannot selectively remove the observed tail. We retain completion and require the independent repair-or-abstain head for Actor-level authority.

The exact-once fresh cohort independently preserves this boundary: only 62.72% add no visible violation, 2.29% have exact zero visible contradiction, and 19.50% worsen in Chamfer. Among the latter, just 16.67% retain a non-inferior visibility F-score. Fresh confirmation therefore strengthens pooled three-dimensional evidence while continuing to reject a universal Actor certificate.

**Frozen authority–certificate alignment.** An exact identity join to the frozen P4 decisions shows why abstention must name its target. At 77.25% coverage, selected Chamfer-worsening falls from 19.50% to 14.60%, yet selected visible-failure risk is 36.39% versus 37.28% for always-repair. Its one-sided 95% Wilson upper bound is 40.40%, which remains above the baseline point risk; safe-visible AUROC is 0.533, and abstention captures 24.62% of visible failures. P4 authorizes Chamfer-nonworsening re-

Table 1. Visibility-conditioned target-only evidence on consumed and independently frozen fresh AV2 cohorts. Higher hit/precision/F-score is better; lower early termination is better. Each audit was frozen before its metrics were read.

| Surface                     | Target-hit recall (%) | Early termination (%) | Visible precision (%) | Visibility F-score |
|-----------------------------|-----------------------|-----------------------|-----------------------|--------------------|
| Consumed clean query        | 49.69                 | 3.395                 | 99.61                 | 0.663              |
| Consumed corrupted baseline | 55.61                 | 3.589                 | 98.38                 | 0.711              |
| Consumed HARP-3D            | 69.83                 | 2.868                 | 99.77                 | 0.822              |
| Fresh clean query           | 49.72                 | 3.448                 | 99.51                 | 0.663              |
| Fresh HARP-3D               | 68.99                 | 2.931                 | 99.68                 | 0.815              |

076 pair but provides no finite-cohort confidence separation for  
077 bidirectional visibility.

078 **Why a stricter provenance rule fails.** If authority ad-  
079 ditionally requires zero COMPLETE outputs, every re-  
080 tained point has a KEEP or matched-hit PROJECT wit-  
081 ness. Provenance-only coverage is 36.90%; after conjunc-  
082 tion with P4 it is 23.71%. Dual visible failure falls to  
083 20.16% (one-sided 95% upper 26.70%), but Chamfer wors-  
084 ening rises to 43.55% and hazardous-Actor coverage is  
085 only 3.52%. Completion-count unsafe-visible AUROC is  
086 0.589. Thus observed provenance improves one physical  
087 axis while failing joint geometry and hazard authority; it  
088 cannot certify unseen views.

089 **A targeted visibility head still fails joint authority.** A  
090 frozen nuScenes-only head reaches safe-visible AUROC  
091 0.753 on source test and 0.625 on AV2. Visibility-only  
092 AV2 selection has 11.63% risk at 8.22% coverage. Its P4  
093 conjunction has 10.26% risk (95% upper 20.98%) but only  
094 7.46% coverage, 35.90% Chamfer worsening, and 3.52%  
095 hazard coverage. The ranking transfers; the authority does  
096 not.

## 097 C. Frozen Cohorts and Exact-Once Rules

098 After a quality read, scenes/logs cannot be replaced, failed  
099 cases remain in the denominator, and no threshold, ra-  
100 dius, feature group, or checkpoint may change. Down-  
101 load/resource failures are reported separately from scientific  
102 failures. The fresh AV2 protocol is bound to the already  
103 frozen P6-C/P4 artifacts and is not changed by the rejected  
104 fresh nuScenes result.

105 The fresh AV2 read contains 523 Actors. P6-C/P4 re-  
106 pair AUROC is 67.62/65.48, coverage is 83.94/77.25%,  
107 and selective Chamfer is 0.178/0.184 m. This external re-  
108 versal does not dominate P4: false repair is 12.43% ver-  
109 sus 11.28%, and P6-C independently loses fresh-nuScenes  
110 AUROC and AURC. The complete evidence therefore re-  
111 tains P4 and rejects a domain-invariant sparsity-consistency  
112 claim.

## D. Actor-Level Decision Intervals

For the frozen ReLU validity MLP, an input box  $[\ell, u]$  is  
propagated through each affine layer using

$$\ell' = W^+ \ell + W^- u + b, \quad u' = W^+ u + W^- \ell + b, \quad (3)$$

followed by endpoint-wise ReLU. Here  $W^+ = \max(W, 0)$   
and  $W^- = \min(W, 0)$ . The resulting logit interval is  
mapped through the sigmoid. An Actor is *robust-select* if its  
lower probability is above the frozen P4 threshold, *robust-*  
*abstain* if its upper probability is below threshold, and unre-  
solved otherwise. This is deterministic interval-bound prop-  
agation over a specified box, related to robust-OOD certifi-  
cation [1]; it does not assert that the box describes an unseen  
sensor.

We fix radii 0.05/0.10/0.20 in nuScenes-training stan-  
dard deviations. The sensor-opportunity group contains  
point/surfel counts, temporal/view support, range, and  
frame count; the physical-surface group contains surface  
residuals, keep/unknown fractions, completion fraction, and  
compiled/query ratio. Their union is exactly the 13D valid-  
ity input.

At radius 0.10, 33.33% of nominal nuScenes-test selec-  
tions and 91.23% of consumed-AV2 selections are robust-  
select. Nevertheless, 10.76% of the latter are retained target  
false repairs. Query-point count is the largest one-feature  
interval source for 83.33%/95.11% of nuScenes/AV2 Ac-  
tors. Thus a shifted model can be confidently and locally  
stable while wrong.

Table 2. Cohort construction is metadata-only. No quality, model score, RGB appearance, or target metric is read before freezing.

| Cohort                    | Size      | Frozen rule                                                                                                                                         |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| AV2 consumed external     | 30 logs   | Sort all 150 Sensor-val UUIDs; take indices 0, 5, . . . , 145. First 20 are quantitative and final 10 qualitative.                                  |
| AV2 fresh exact-once      | 20 logs   | Remove the consumed indices; from the 120-log complement take positions 0, 6, . . . , 114.                                                          |
| nuScenes fresh exact-once | 20 scenes | Exclude every P4 train/calibration/test scene; sort the remaining 106 available scenes by official index and take rounded equally spaced positions. |

## E. Failure Ledger and Claim Corrections

Table 3. Paper-facing failures are retained as boundaries rather than hidden by replacement experiments.

| ID     | Observation                                                                                 | Resulting boundary or correction                                                                                                             |
|--------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| V7-F09 | Aggregate geometry hides 17.19% Actor-level Chamfer worsening.                              | P3 is aggregate hard evidence; P4 adds repair-or-abstain and preserves all Actors.                                                           |
| V7-F11 | AV2 validity attribution is dominated by sensor opportunity.                                | Zero hazard leakage does not imply domain-invariant validity.                                                                                |
| V7-F13 | Exact P4-train overlap is only two scenes/five Actors.                                      | Reject joint fitting; audit the frozen physical/reliability interface only.                                                                  |
| V7-F14 | FP32 exceeds the analytic tolerance by $3.8 \times 10^{-6}$ .                               | Recompute the same bound in FP64; do not loosen the tolerance or alter rows.                                                                 |
| V7-F15 | Fresh P6-C repair AUROC loses 3.50 points to P4.                                            | Reject recovery despite higher operating-point coverage; P4 remains primary.                                                                 |
| V7-F16 | Freeze prose mislabeled the P346 held-out horizon.                                          | Executable artifact is 2.5 s; correct documentation without rerunning.                                                                       |
| V7-F17 | 47 AV2 false repairs remain robust-select at radius 0.10.                                   | Network stability is not correctness, calibration, or a safety certificate.                                                                  |
| V7-F18 | Pooled ray physics improves, but consumed/fresh nonnew-visible rates are only 64.04/62.72%. | Retain aggregate C1 evidence and the completion gain-tail trade-off; reject a universal Actor certificate and do not tune the ray tolerance. |
| V7-F19 | P6-C loses fresh-nuScenes ranking but improves fresh-AV2 AUROC while raising false repair.  | Report the domain reversal; an external gate pass neither overrides the source rejection nor promotes P6-C over P4.                          |

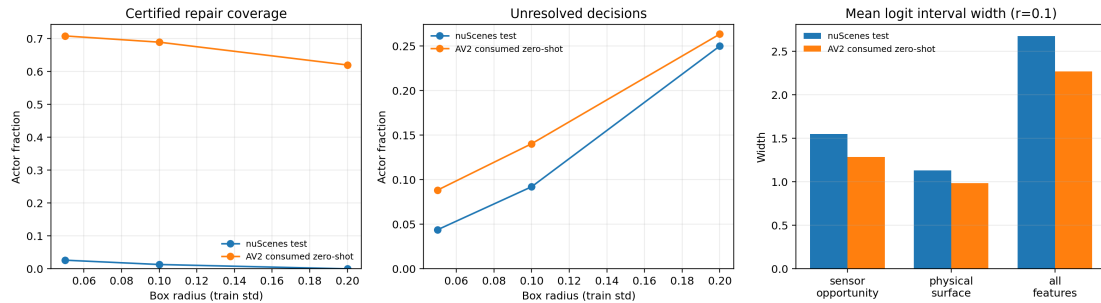


Figure 1. Frozen P7-C interval audit. Left/center: robust-select coverage and unresolved fraction across pre-registered radii. Right: sensor-opportunity boxes induce wider logit intervals than physical-surface boxes. The high AV2 certified coverage does not erase the 47 stable false repairs.

## F. Frozen AV2 Camera-Evidence Gallery

The following ten cases are the actor-rank-0 case from each qualitative log in the pre-registered order, not a metric or appearance ranking. Every panel shows synchronized RGB, the paired synthetic artifact probe, the four-action physical output, and sparse camera depth. The full package contains 30 panels and 30 before/after videos (three lexically first eligible Actors per log). Videos diagnose temporal behavior but are not photorealistic renders.



Figure 2. Frozen case q00, hazardous Actor 1f197615, ring-front-center; 251 visible query returns and 2,287 sparse-depth points in crop.



Figure 3. Frozen case q01, non-hazardous Actor 0e0562b6, ring-rear-left; 137 visible query returns and 1,770 sparse-depth points.



Figure 4. Frozen case q02, non-hazardous Actor 08fc423e, ring-side-right; 85 visible query returns and 2,476 sparse-depth points.





Figure 5. Frozen case q03, hazardous Actor 05cccb41, ring-side-left; 2,456 visible query returns and 17,286 sparse-depth points.

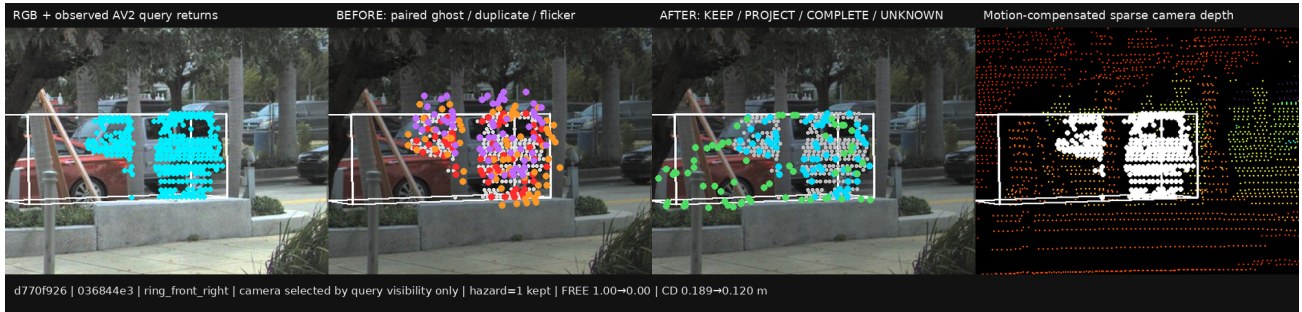


Figure 6. Frozen case q04, hazardous Actor 036844e3, ring-front-right; 292 visible query returns and 2,358 sparse-depth points.

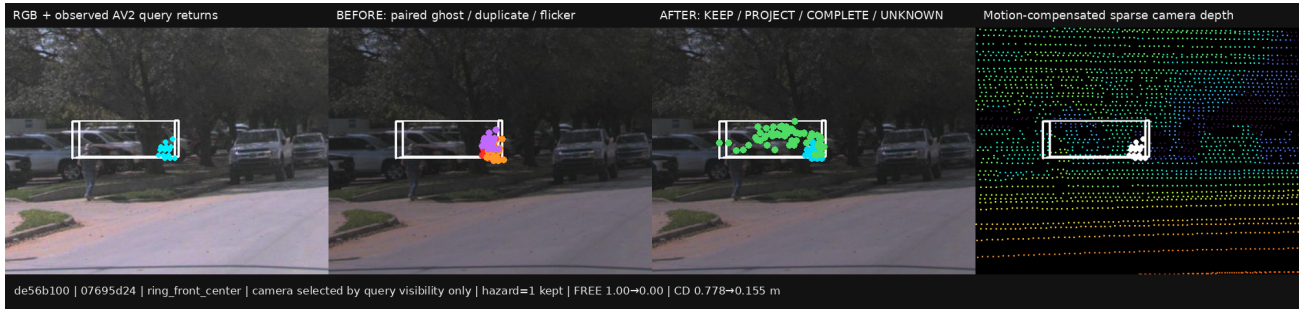


Figure 7. Frozen case q05, hazardous far/sparse Actor 07695d24, ring-front-center; all 14 query returns remain visible.



Figure 8. Frozen case q06, non-hazardous Actor 08515e0d, ring-front-right; 79 visible query returns and 2,757 sparse-depth points.



Figure 9. Frozen case q07, non-hazardous Actor 076c9103, ring-front-center. This unfiltered case exposes Chamfer worsening and is retained.

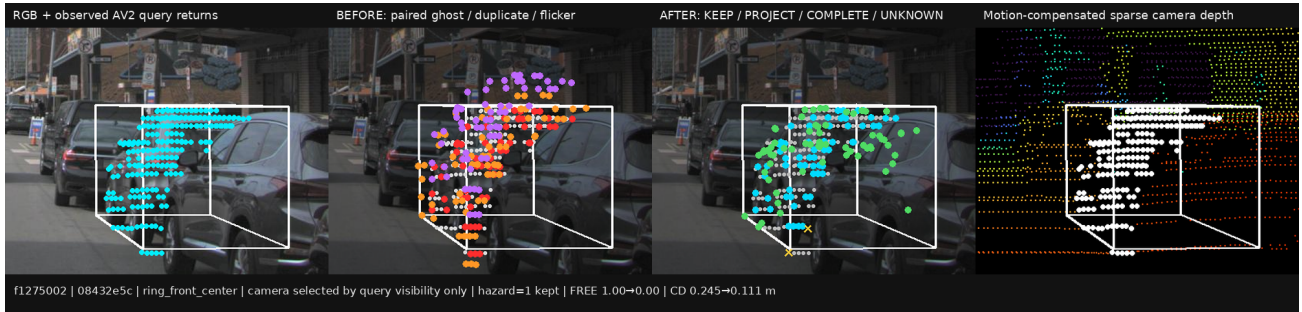


Figure 10. Frozen case q08, hazardous Actor 08432e5c, ring-front-center; 231 visible query returns and 2,101 sparse-depth points.



Figure 11. Frozen case q09, non-hazardous Actor 6d55ff95, ring-front-center; 38 visible query returns and 1,630 sparse-depth points.

## 146 G. Resource and Reproducibility Boundary

147 All V7 scientific runs use one RTX 3090. The largest paper-facing GPU allocation is below 1.5 GiB, while P7-C uses 9.2 MiB  
148 and 1.37 s. Download and preprocessing are sequential on the low-memory host. Artifacts are identified by task/run paths  
149 and immutable cohort/config files; the protocol intentionally does not add hashes, checksums, fingerprints, repeated smoke  
150 matrices, or post-hoc quality gates.

## 151 References

- 152 [1] Julian Bitterwolf, Alexander Meinke, and Matthias Hein. Certifiably adversarially robust detection of out-of-distribution data. In  
153 *Advances in Neural Information Processing Systems*, 2020. 2
- 154 [2] Peiyun Hu, Jason Ziglar, David Held, and Deva Ramanan. What you see is what you get: Exploiting visibility for 3d object detection.  
155 In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11001–11009, 2020. 1